



BICYCLE SAFETY ROUTING FOR BERLIN

Capstone project · GIS baseline + Leakage-safe ML +
Severity evidence

1 • Business Problem And Product

WHAT PROBLEM DOES 2W1C SOLVE?

CYCLIST PAIN POINT

Navigation apps optimize distance and time very well, but rarely explain route safety.

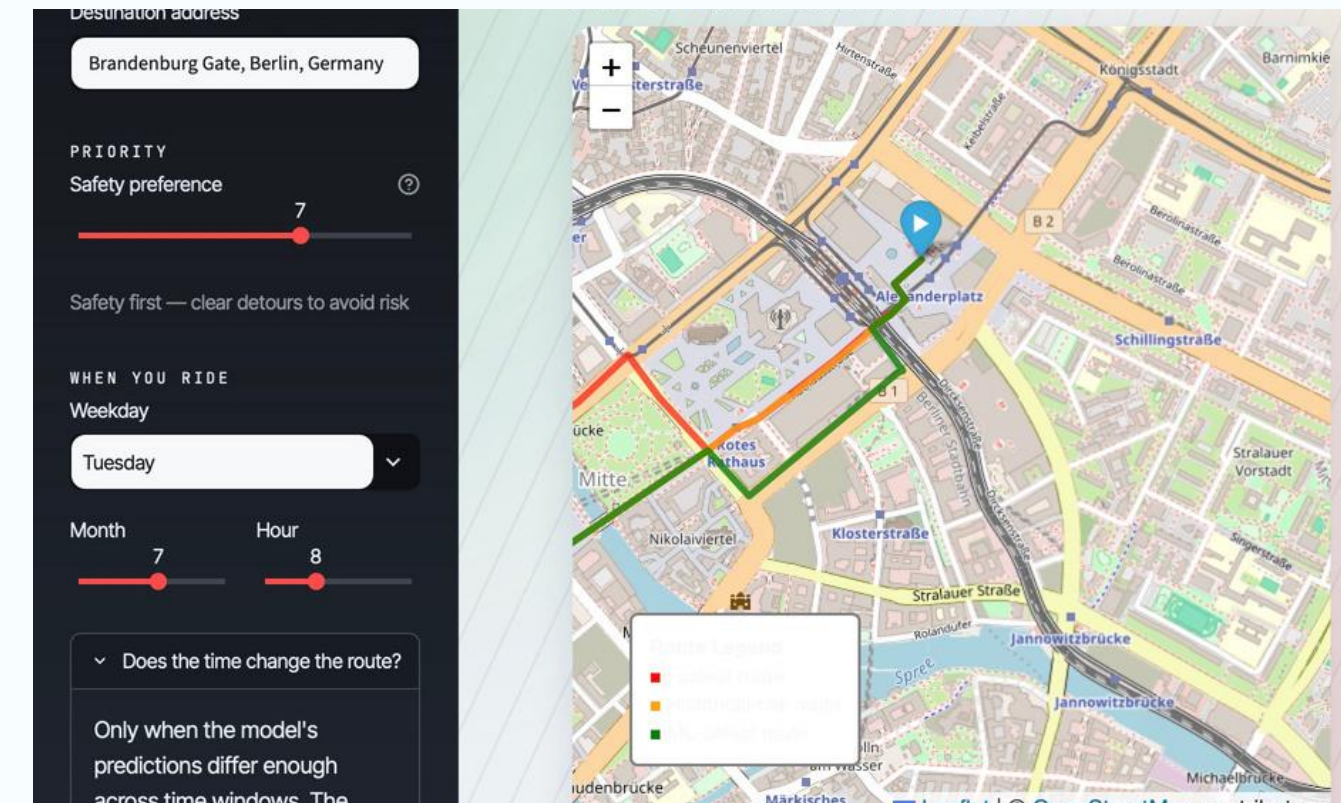
For a cyclist, the useful question is not “where will the **next crash** happen?”

but “**which route avoids historically risky street** segments with a reasonable detour?”

BUSINESS VALUE

For city mobility products, bicycle insurance, logistics, or public-sector planning, the value is interpretable risk-aware routing.

It is also auditable: the app shows where risk comes from, what the model can prove, and what it cannot prove.



PRODUCT PROMISE

2W1C compares three choices:

1. Fastest route
2. Historical GIS-risk route
3. Leakage-safe ML road-risk route

The output is a map plus the risk-distance trade-off.



3 · Dataset : Origin, Quality And EDA

Unfallatlas + OpenStreetMap, validated before modelling

CLEAN
CRASHES

37,948

Berlin bicycle crashes,
2018–2025

SNAPPED TO
OSM

37,896

99.9% within 25 m

MEDIAN
OFFSET

0.7

High coordinate
quality

NEAR
JUNCTION

80.8%

within 20 m of node

Origin

Unfallatlas crash records

OSM bicycle road graph

Edge features: road class
length, cycleway, speed, limit

Quality
Checks

25 m **snapping threshold**

Junction Distance calculated

Raw columns kept for **severity
evidence**

EDA
Lesson

Crashes are spatially
clustered

Many are close to **intersections**

Time patterns exist in **EDA**

4 • Baseline model

Historical GIS risk is the auditable product baseline

Selected baseline

Historical GIS-risk route

Why this baseline?

- transparent
- easy to explain
- works without label leakage
- directly maps past crashes to street segments
- gives a user-visible route comparison

How it works

1. Snap crash points to OSM edges
2. Weight severity
3. Aggregate by undirected street segment
4. Add junction risk
5. Penalize high-risk edges in routing cost

routing cost
 $\text{length} \times (1 + \text{safety preference} \times \text{risk}) + \text{junction penalty}$

Forward validated
Built on 2018–2023, tested on 2024–25: the riskiest 22% of network length contains 55% of future crashes — 2.56× lift over random.

Product output
59% lower risk exposure for a 0.89 km detour — not “the objectively safest route”.

5 • Models and leakage audit

A better product came from diagnosing the too-good model

Model	Status	PR-AUC	Lift	Meaning
time_only	diagnostic	0.4125	1.00×	matches base rate by construction: negatives inherit positive timestamps
road_only	DEPLOYED	0.7228	1.76×	best leakage-safe product model
road + time	diagnostic	0.7049	1.71×	time does not improve the road model
leaky	INVALID	0.8892	2.16×	uses accident-derived columns; not reported as performance

Leakage removed
Every positive had accident_count ≥ 1 by construction; only 7.6% of negatives did. Recall hit exactly 1.000 — the model read the label.

Negative sampling fixed
Negatives restricted to rideable road classes, so the task is not “road versus footpath”.

Honest result
0.72 against a 0.41 base rate. Lower than 0.89, and the only figure we can defend.

6 • Metric choice

Why PR-AUC and lift are more useful than accuracy

Selected metrics

Primary: PR-AUC
Secondary: lift over base rate, top-10% recall, Brier score
Product: forward-validation lift, then risk reduction vs detour

Why PR-AUC?

The positive class is rare and the product is a risk-ranking problem. We care whether high-risk segments appear near the top of the ranking, not just overall accuracy.

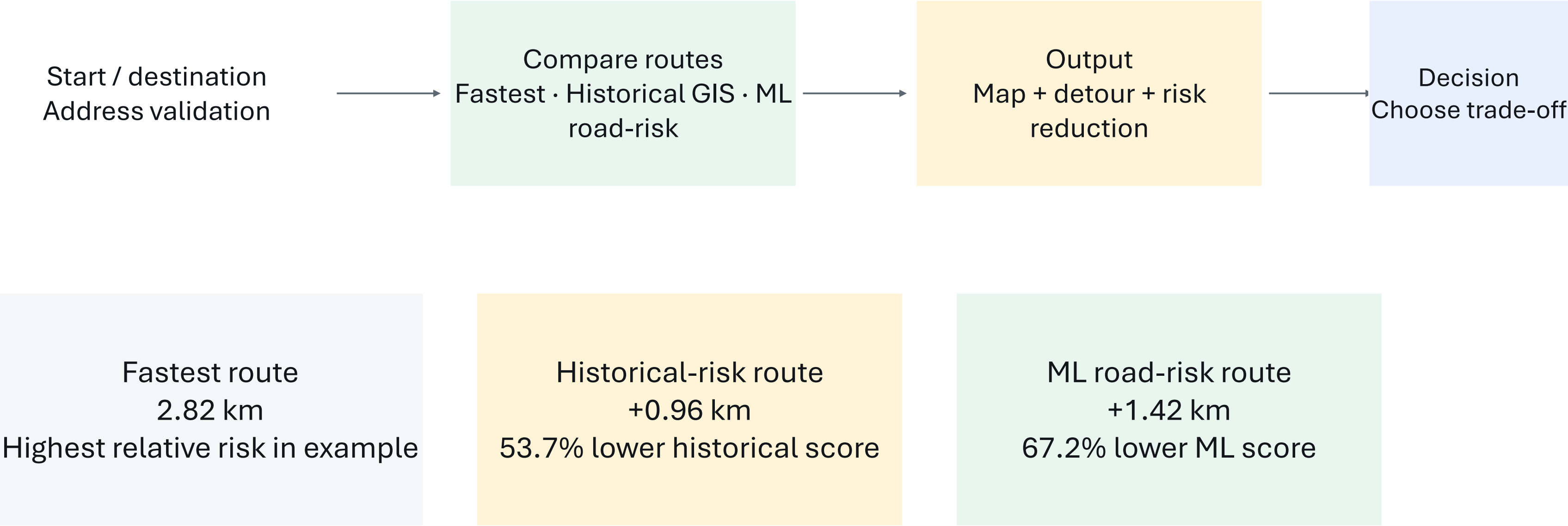
What not to use alone

Accuracy can look good even when the model misses risky segments. ROC-AUC is useful, but PR-AUC is more informative when positives are relatively scarce.



7 • Product demo flow

What the user sees in the app



The percentages are lower model scores, not lower real-world crash probability.

2W1C

Two wheels, one city — Berlin

TRIP

Start address

Alexanderplatz, Berlin, Germany

Destination address

Brandenburg Gate, Berlin, Germany

Not checked yet.
Press Enter after typing the address.

Check addresses

PRIORITY

Safety preference

7

Safety first — clear detours to avoid risk

WHEN YOU RIDE

Hour

8

> Does the time change the route?

Compare routes

BERLIN · DEFENSIBLE GIS · ML ROUTE ENGINE

2W1C: Bicycle Safety Routing

Compare the shortest route, a severity-weighted historical GIS-risk route, and a leakage-safe ML road-risk route. The app separates spatial risk, ML diagnostics, and severity evidence instead of hiding model limitations.

Fastest · distance only

Historical · GIS risk baseline

ML · leakage-safe road-only model

Severity · evidence, not a route guarantee

Route comparison

Leakage-safe ML

Historical GIS risk

Severity evidence

Method

FASTEST

2.82 km

distance only — the baseline

historical risk0.6896

ML risk0.7464

segments33

HISTORICAL GIS RISK

3.55 km

+0.73 km vs fastest

risk vs fastest-55.5%

historical risk0.3070

baseline0.6896

segments68

ML ROAD RISK

2.98 km

+0.17 km vs fastest

risk vs fastest-26.8%

ML risk0.5465

baseline0.7464

segments57

Risk values are relative model scores. Historical GIS risk and ML road-risk are reported on their own scales and should not be interpreted as personal crash probabilities.

MAP

Fastest

Historical GIS risk

ML road risk

Moabit

Prezliauer Berg

Kreuzberg

Friedrichshain

Tiergarten

Hansaviertel

Großer Tiergarten

Grünewald

Wilmersdorf

Neukölln

Tempelhof

Marzahn

Hellersdorf

Adlershof

Charlottenburg

Wilmersdorf

Neukölln

Tempelhof

Marzahn

Hellersdorf

Adlershof

Charlottenburg

Legend

Fastest route

Historical GIS-risk route

ML road-risk route

RECOMMENDATION

Route comparison:

- Fastest route: 2.82 km
Historical risk exposure score: 0.6896

- Historical GIS-risk route: 3.55 km, historical risk 0.3070
Detour: 0.73 km; historical risk reduction: 55.5%

ML road-risk evaluation of fastest route: 0.7464

- ML road-risk route: 2.98 km, ML risk 0.5465
Detour: 0.17 km; ML risk reduction: 26.8%

Interpretation: these are relative model scores, not personal crash probabilities. The historical route reduces historical spatial-risk exposure; the ML route uses a leakage-safe road-only occurrence model.

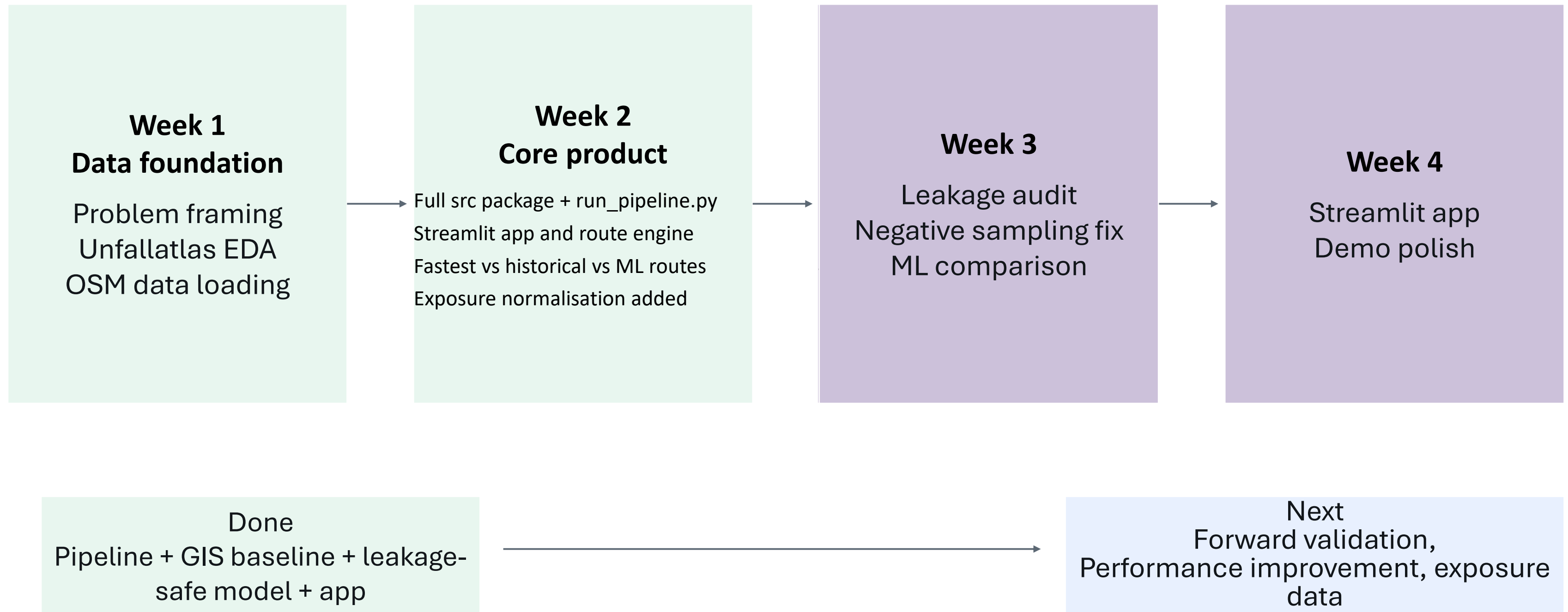
SIDE BY SIDE

	route	distance_km	risk_model	risk_score
0	Fastest	2.8174	distance only	0.6896
1	Historical GIS-risk	3.5491	historical spatial risk	0.3070
2	ML road-risk	2.9842	leakage-safe road-only ML	0.5465

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8 • Timeline and current status

Week 2 status: working pipeline, stronger validation, clearer risks



9 • Future work and difficulties

What would make the prototype stronger and harder to break

Exposure data

Add bicycle-volume data to separate risk from traffic volume.

App speed

Current routing can take >2 min; cache graph costs and precompute risks.

User reports

Let cyclists upload unreported crashes and near misses.

Better ML

Compare more models and ensembles to improve valid PR-AUC.

Time-aware risk

Use hour as severity modifier only when data supports it.

Validation

Test whether past high-risk streets predict future crashes.

- **Improve the product without overclaiming — route scores are relative model evidence, not real-world crash probabilities.**

10 • Takeaway

A working prototype, a stronger model story, and a clear civic motivation

What we built

A Berlin bike-routing MVP
Fastest, historical-risk and ML routes
Address validation + demo-ready app

What we learned

Historical risk predicts future crash concentration
Leakage was found and removed
Risk scores are model evidence, not crash probabilities

Outlook

Make the app faster and easier to use
Add user-reported, unreported crashes
Compare and combine more ML models

Write to us at leser-blz@berlinerverlag.com! In the coming weeks, we will continue to research how Berlin can become more bicycle-friendly.
(alm./who.)



Thank you